

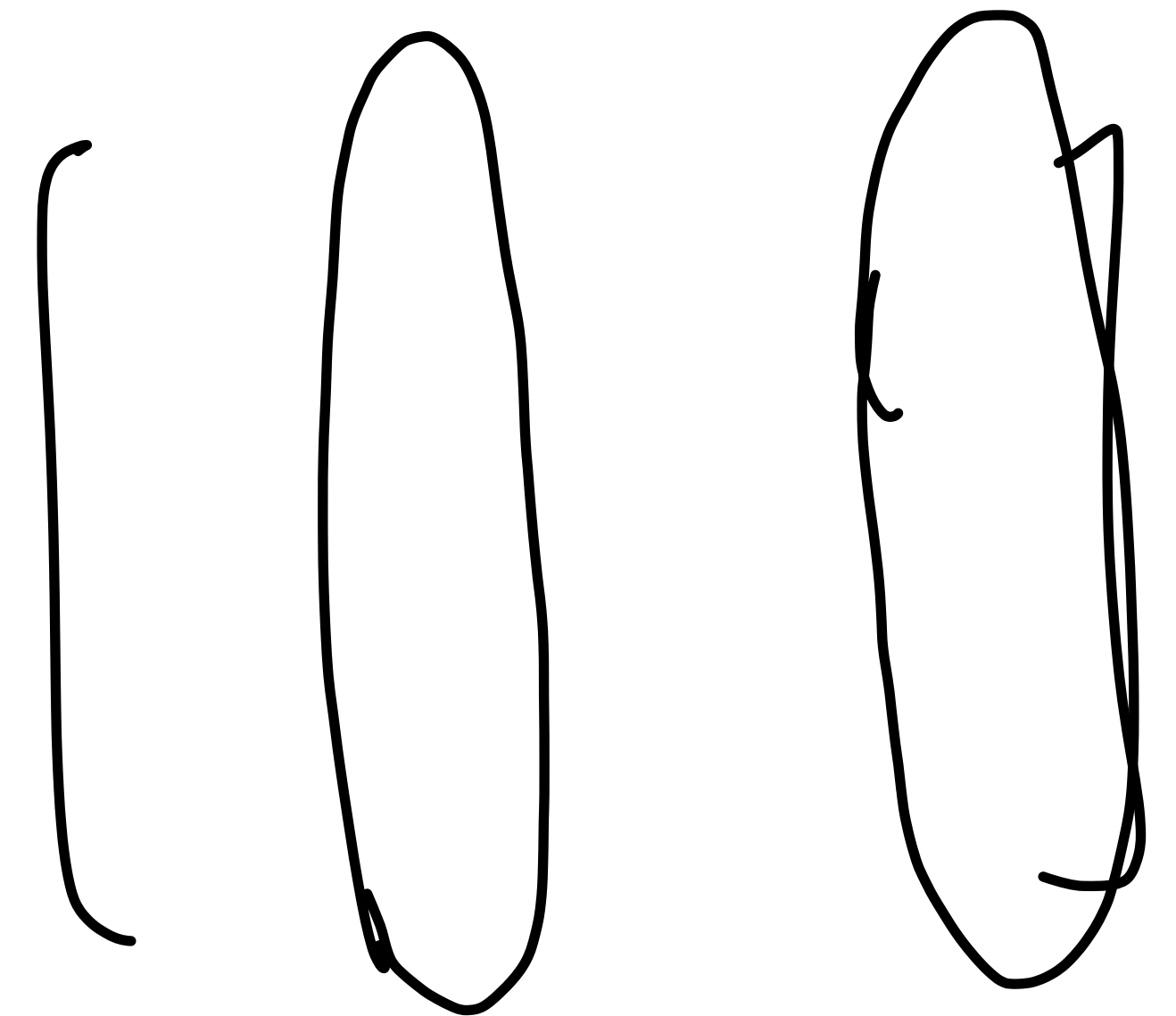
Vector Equations

Geometric Algorithms
Lecture 4

Practice Problem

Suppose that A is a 322×245 augmented matrix for a system with infinitely many solutions. What is the maximum number of pivot positions that A can have?

2 4 5
2 4 3
2 4 4



Answer

$\underbrace{5}_{\text{rows}} \times \underbrace{3}_{\text{column}} \text{ matrix}$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 2 \\ 2 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad 0 = 1$$

A is an aug. mat. for inf. sol. system.

Answer

$$A \sim \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = a$$

$$x_2 = b$$

Answer

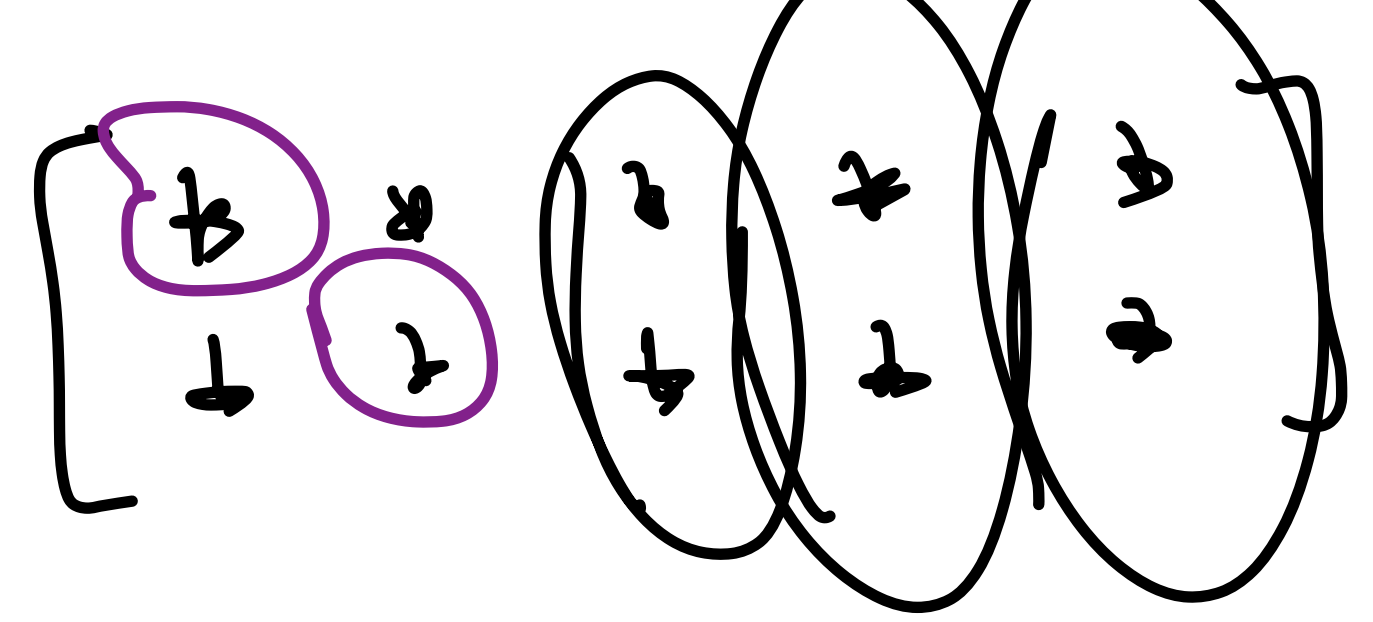
$$A \sim \begin{bmatrix} 1 & a & b \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = b - a x_2$$

x_2 is free

Answer

2×5



A is an $(m \times n)$ matrix s.t.

A is a good mat. for inf. sol. system.

$$\text{max pivot pos} = \min(m, n - 2)$$

$$\begin{bmatrix} \boxed{2} & 2 & 2 & 2 \\ 2 & \boxed{2} & 1 & 1 \\ 2 & 2 & 2 & \boxed{2} \end{bmatrix}$$

$$\begin{bmatrix} \boxed{2} & 4 & 1 & 0 \\ 0 & \boxed{1} & -1 & 1 \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

2

$$\begin{bmatrix} \boxed{1} & 0 & 2 & 0 \\ 0 & \boxed{1} & 2 & 0 \\ 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Outline

- » Formally define vectors
- » Discuss vector operations and vector algebra
- » Draw the connection between vectors and systems of linear equations

Keywords

vector

vector addition

vector scaling/multiplication

the zero vector

vector equations

linear combinations

span

A Bit of Motivation

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Data are ordered collections of values

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Data are ordered collections of values

Q: When are two data *similar*?

A Bit of Motivation

Data are ordered collections of values

Q: When are two data *similar*?

A: When they're **close together** geometrically

A Note on Algebra

$$\mathbf{v} = \mathbf{w}$$

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

A Note on Algebra

In programming, an *interface* is an abstract collection of related functions

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In programming, an *interface* is an abstract collection of related functions

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And object then *implements* an interface

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A Note on Algebra

In programming, an *interface* is an abstract collection of related functions

$$\mathbf{v} = \mathbf{w}$$

And object then *implements* an interface

$$\mathbf{v} + \mathbf{w}$$

$$a\mathbf{v}$$

Doing abstract algebra is like implementing an *algebraic* interface

Vectors

What is a vector (in $\mathbb{R}^n$)?

A. an n -tuple of real numbers

B. a point in $\mathbb{R}^n$

C. a 1-column matrix with real values

D. all of the above

E. none of the above?

Column Vectors

Df. a *column vector* is a matrix with a single column

$$\begin{bmatrix} 2 \\ 3 \\ 5 \\ \pi \end{bmatrix}$$

Matrix Size

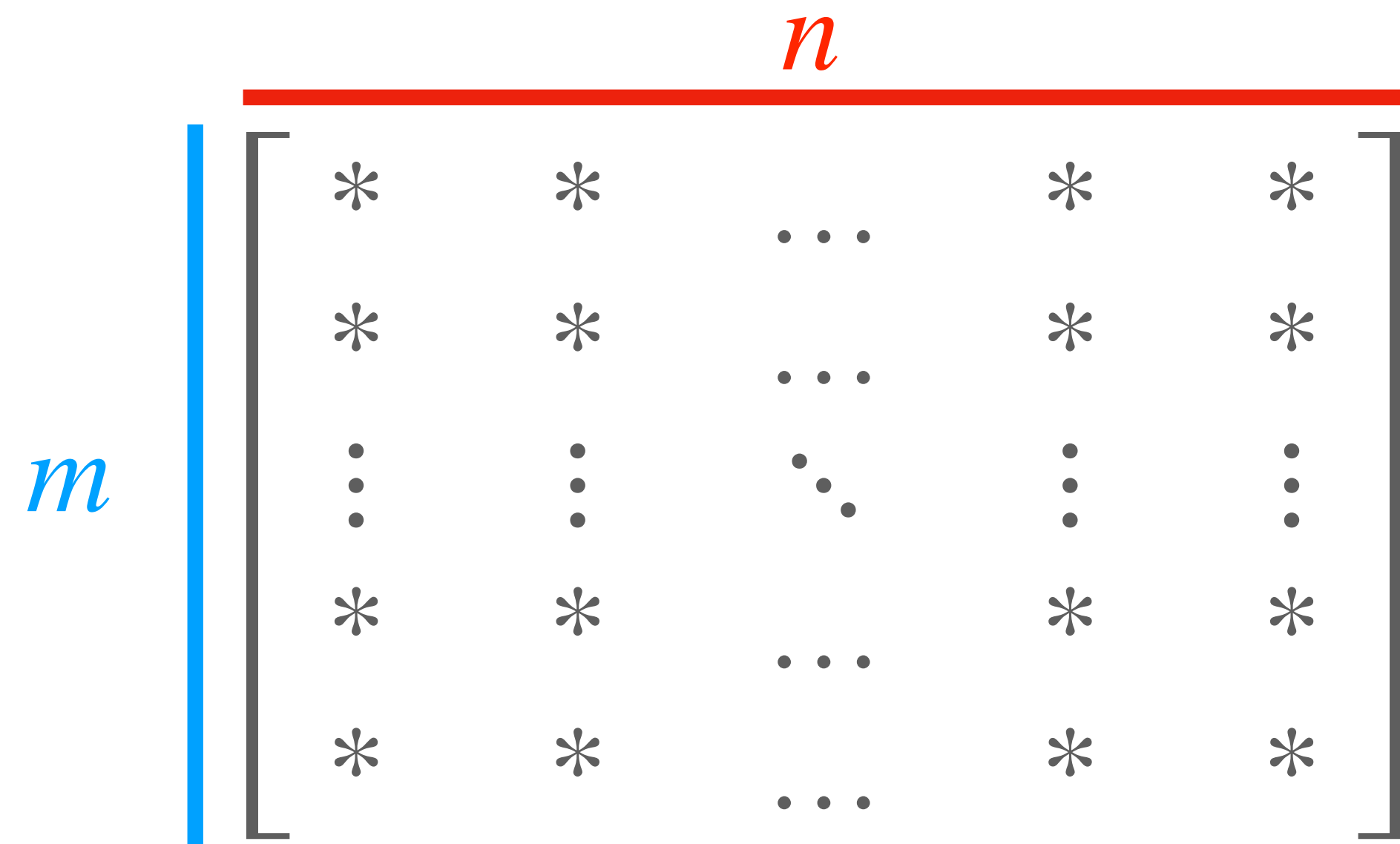
An $(m \times n)$ matrix has m rows and n columns

$$\begin{array}{c} m \end{array} \left[\begin{array}{ccccc} * & * & \dots & * & * \\ * & * & \dots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ * & * & \dots & * & * \\ * & * & \dots & * & * \end{array} \right] \begin{array}{c} n \end{array}$$

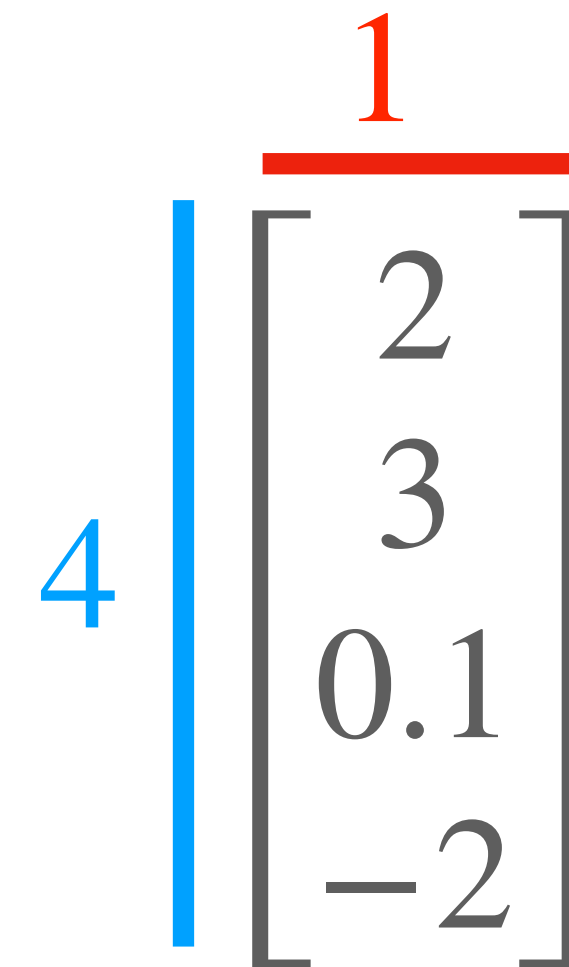
$$\begin{array}{c} 4 \end{array} \left[\begin{array}{c} 2 \\ 3 \\ 0.1 \\ -2 \end{array} \right] \begin{array}{c} 1 \end{array}$$

Matrix Size

An $(m \times n)$ matrix has m rows and n columns



A diagram of a general $m \times n$ matrix. A blue vertical line to the left of the matrix is labeled with a blue italic m . A red horizontal line above the matrix is labeled with a red italic n . The matrix is enclosed in large square brackets and contains five rows and five columns of entries. The entries are represented by asterisks (*), ellipses (...), and a diagonal ellipsis (\(\dots\)) in the third row and third column.



A diagram of a specific 4×1 column vector. A blue vertical line to the left of the vector is labeled with a blue number 4. A red horizontal line above the vector is labeled with a red number 1. The vector is enclosed in large square brackets and contains four entries: 2, 3, 0.1, and -2.

$\mathbb{R}^{m \times n}$ is set of matrices with $\mathbb{R}$ entries

Matrix Size

An $(m \times n)$ matrix has m rows and n columns

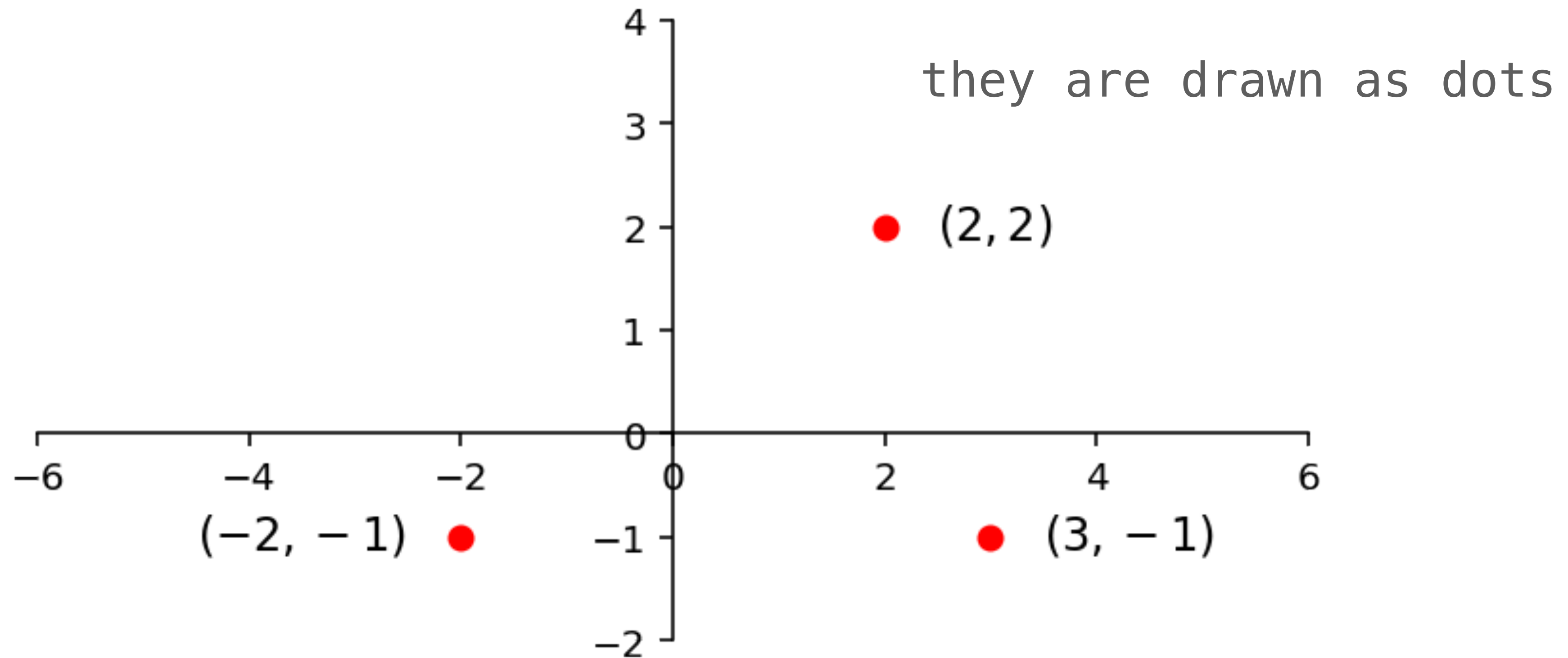
$$\begin{matrix} & & & n \\ & & & \hline m & \left[\begin{array}{ccccc} * & * & \dots & * & * \\ * & * & \dots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ * & * & \dots & * & * \\ * & * & \dots & * & * \end{array} \right] \end{matrix}$$

$$\begin{matrix} & & & 1 \\ & & & \hline 4 & \left[\begin{array}{c} 2 \\ 3 \\ 0.1 \\ -2 \end{array} \right] \end{matrix} \in \mathbb{R}^4$$

the number of rows
of a vectors is
called its **dimension**

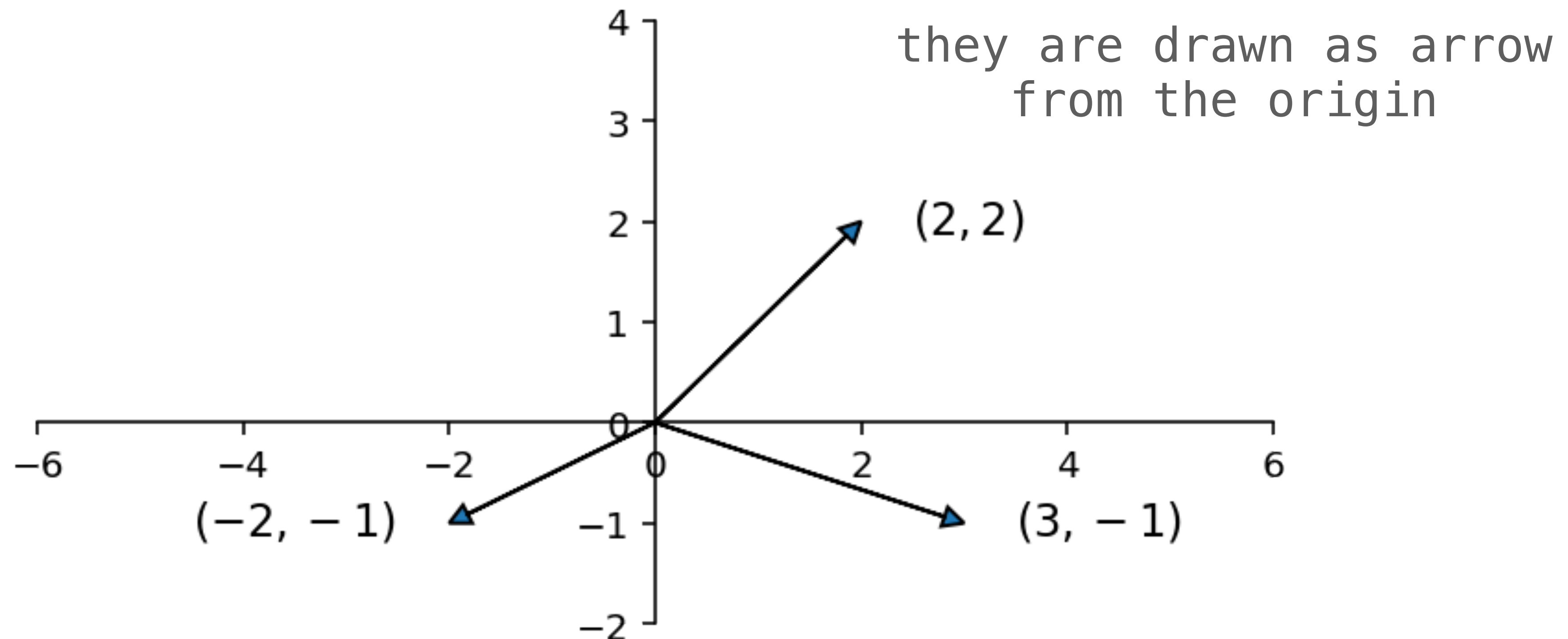
$\mathbb{R}^{m \times n}$ is set of matrices with $\mathbb{R}$ entries

Notation (Points)



points in $\mathbb{R}^2$ are notated as (a, b)

Notation (Vectors)



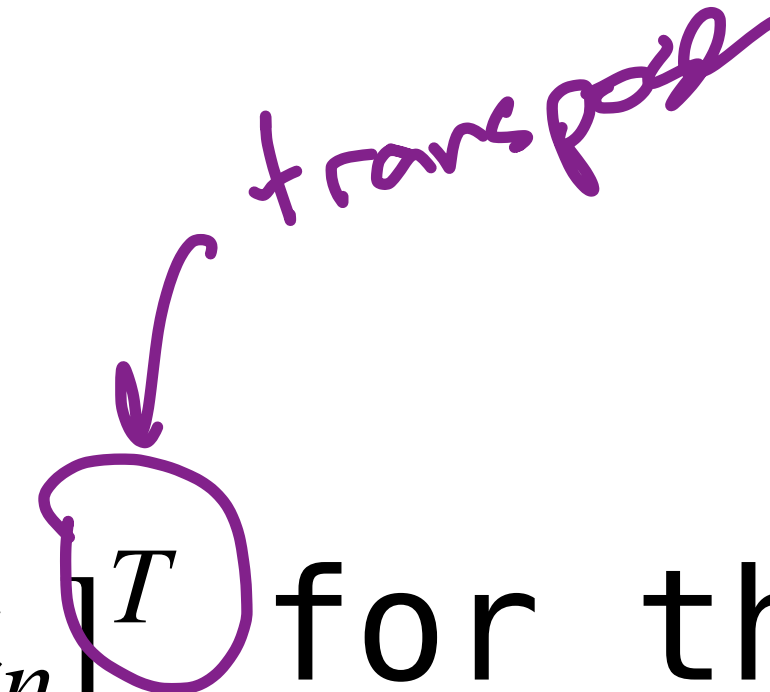
vectors in $\mathbb{R}^2$ are notated as $\begin{bmatrix} a \\ b \end{bmatrix}$

Notation (Looking ahead)

We will often write $[a_1 \ a_2 \ \dots \ a_n]^T$ for the vector $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$

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We will often write $[a_1 \ a_2 \ \dots \ a_n]^T$ for the vector $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$



$$\mathbb{R}^{1 \times n} \neq \mathbb{R}^n$$

IMPORTANT

$(a_1, a_2, \dots, a_n)$ is not the same as $[a_1 \ a_2 \ \dots \ a_n]$



Vector Operations

Vector "Interface"

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equality what does it mean for two vectors
to be equal?

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What properties do they need to satisfy?

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Vector Equality

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

~~$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$~~

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

must be
the same

$$\begin{aligned} x_1 &= y_1 \\ \text{and} \\ x_2 &= y_2 \\ \text{and} \\ &\vdots \\ x_n &= y_n \end{aligned}$$

Vector Equality

two vectors are equal if their entries at each position are equal

(this is also the case for matrices)

!!IMPORTANT!!
ORDER MATTERS

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Vector Equality

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \text{is the same as} \quad \begin{array}{l} a_1 = b_1 \\ a_2 = b_2 \\ \vdots \\ a_n = b_n \end{array}$$

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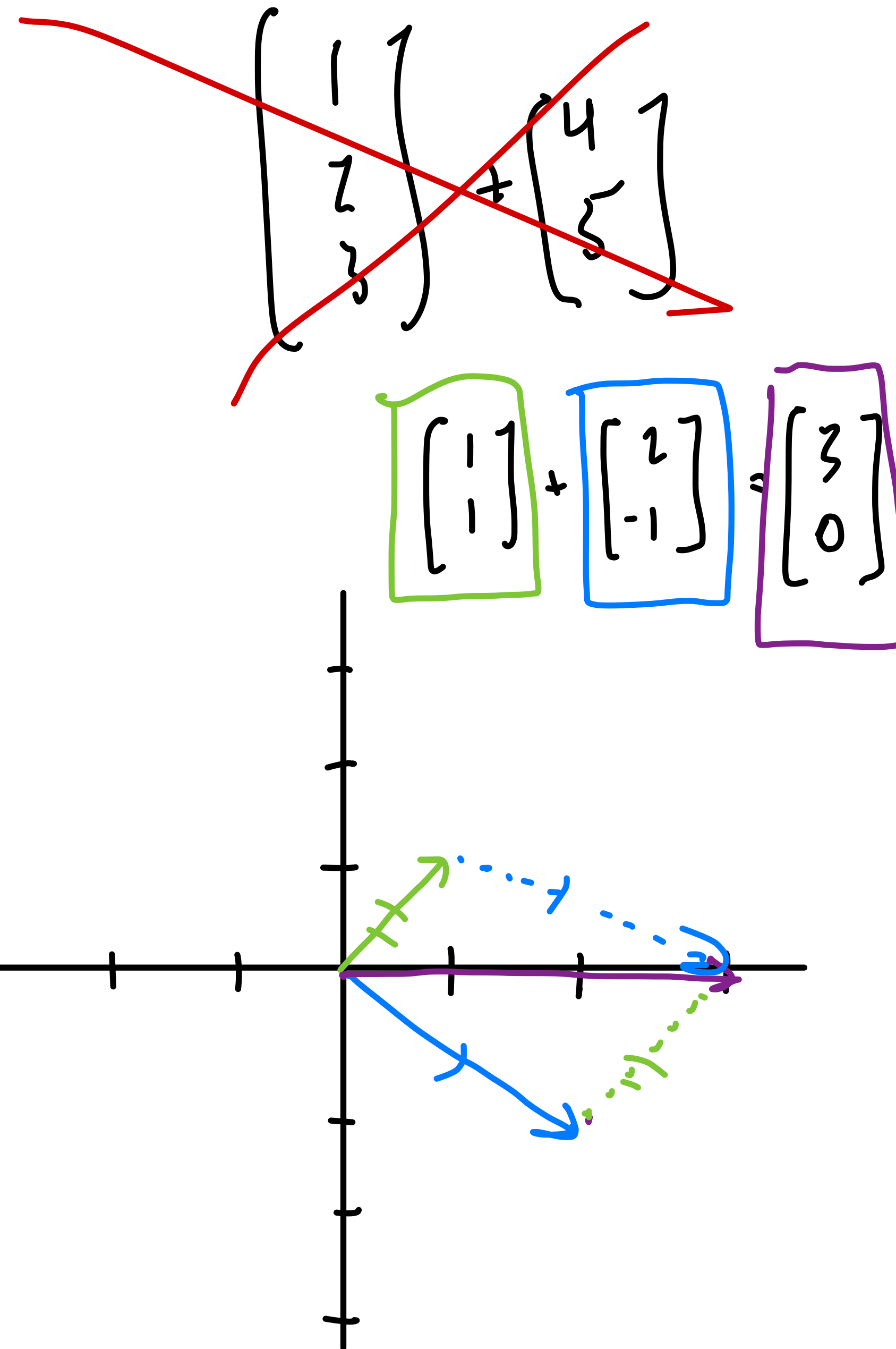
What properties do they need to satisfy?

Vector Addition

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1+3 \\ 2+4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

must be same



Vector Addition

adding two vectors means adding their entries

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}$$

Vector Addition

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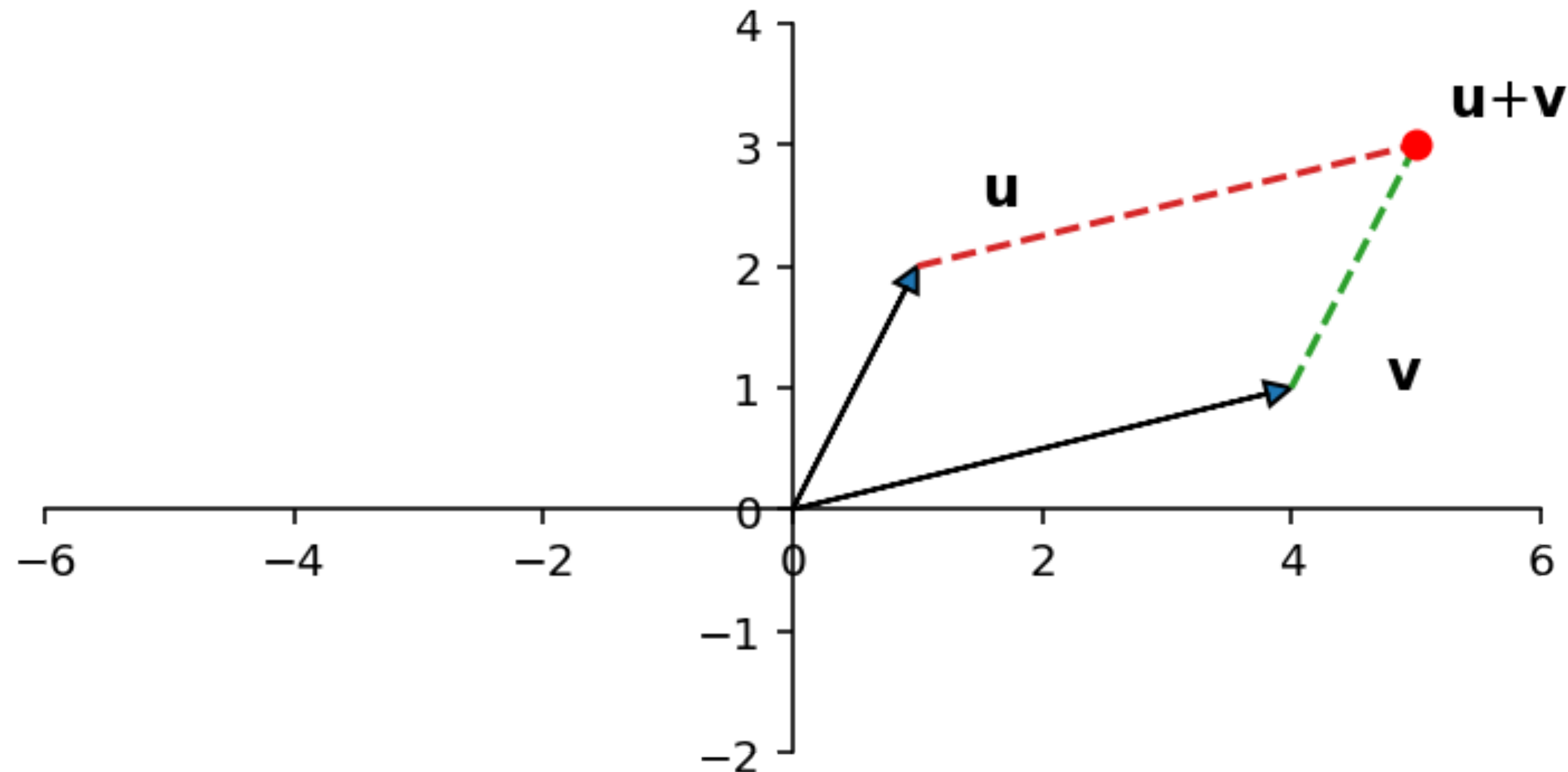
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!! IMPORTANT !!

WE CAN ONLY ADD VECTORS OF THE SAME SIZE

Vector Addition (Geometrically)

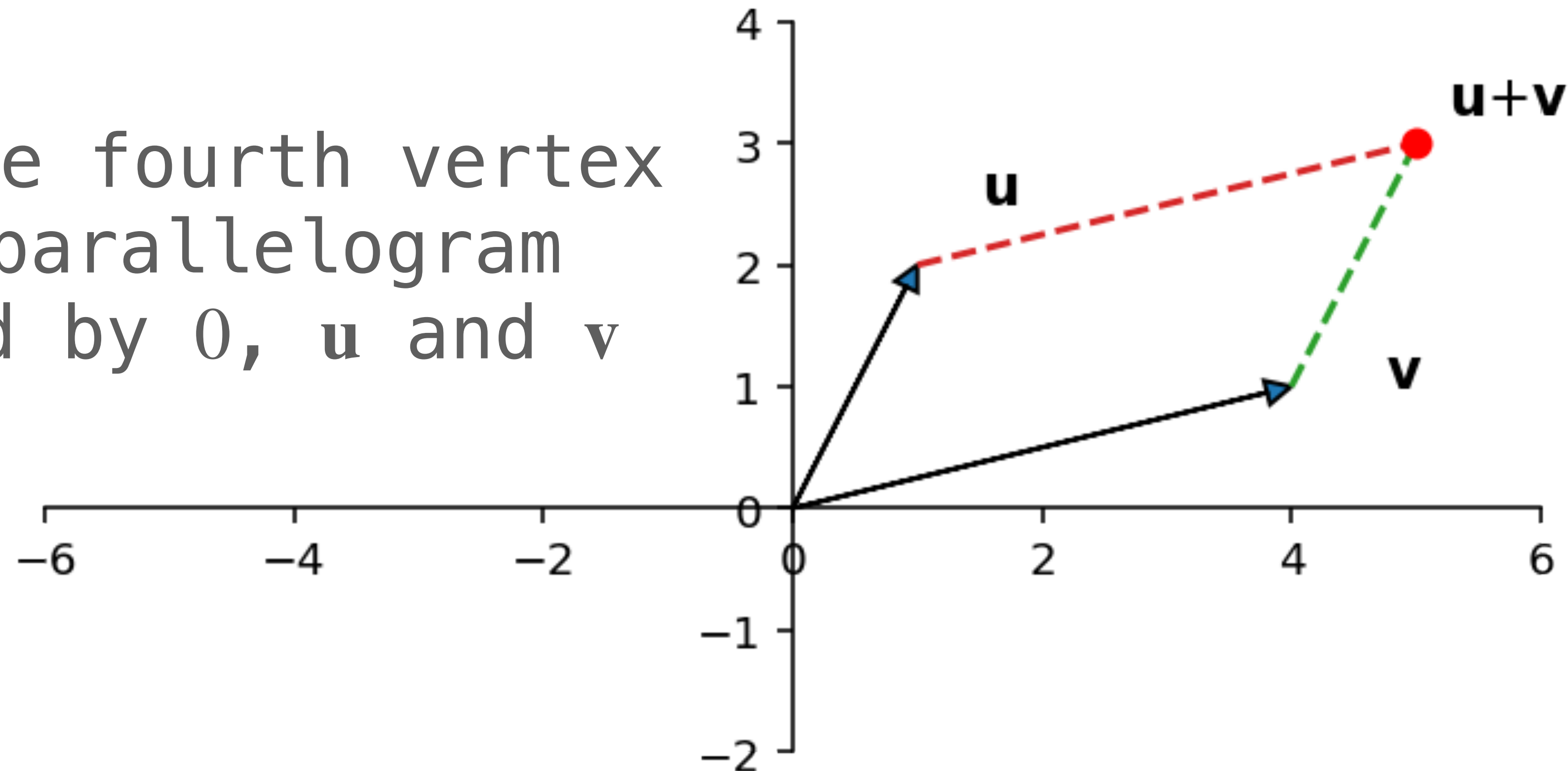
in $\mathbb{R}^2$ it's called the *parallelogram rule*



Vector Addition (Geometrically)

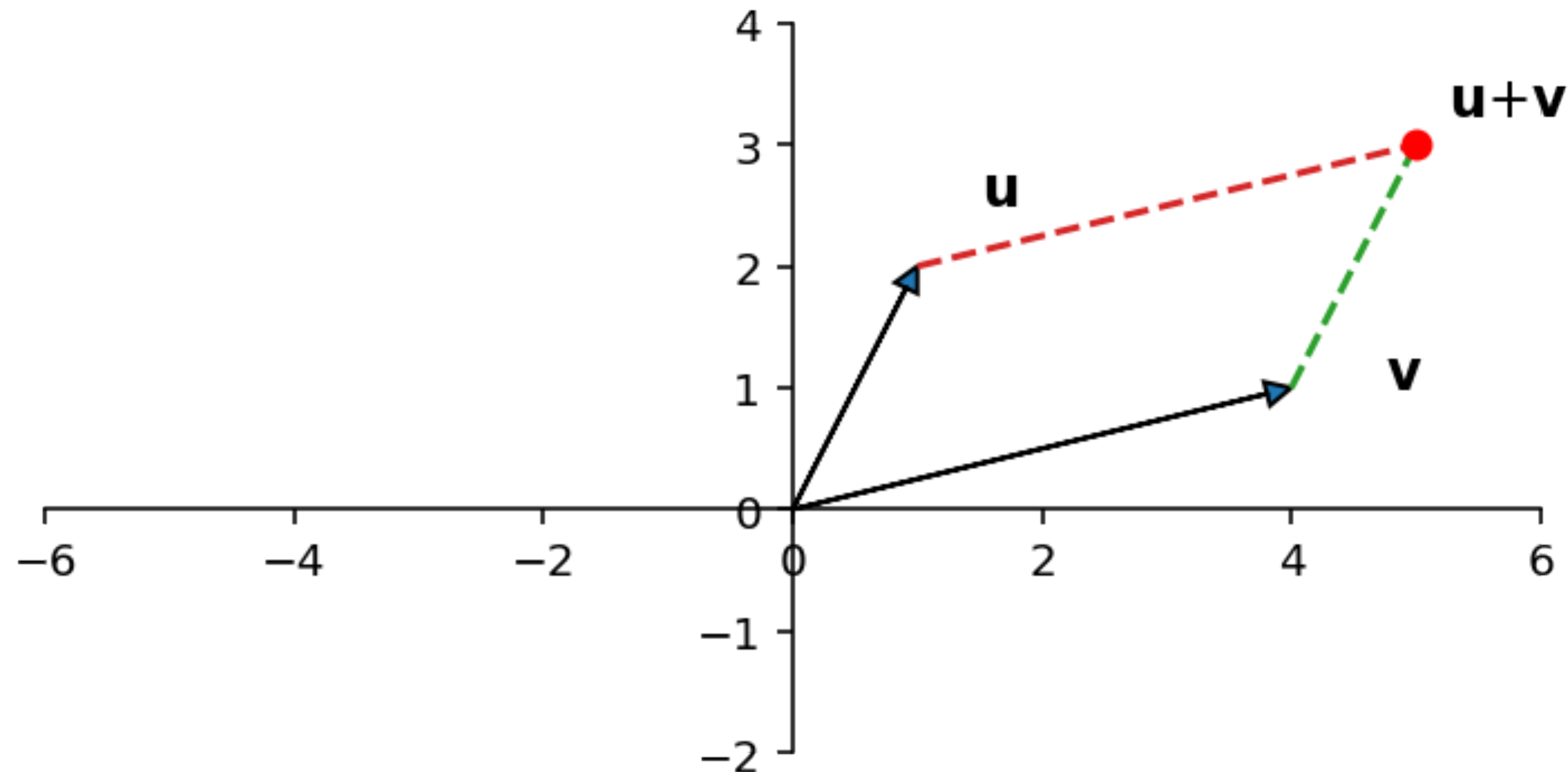
in $\mathbb{R}^2$ it's called the *parallelogram rule*

$\mathbf{u} + \mathbf{v}$ is the fourth vertex
of the parallelogram
generated by $\mathbf{0}$, $\mathbf{u}$ and $\mathbf{v}$



Vector Addition (Geometrically)

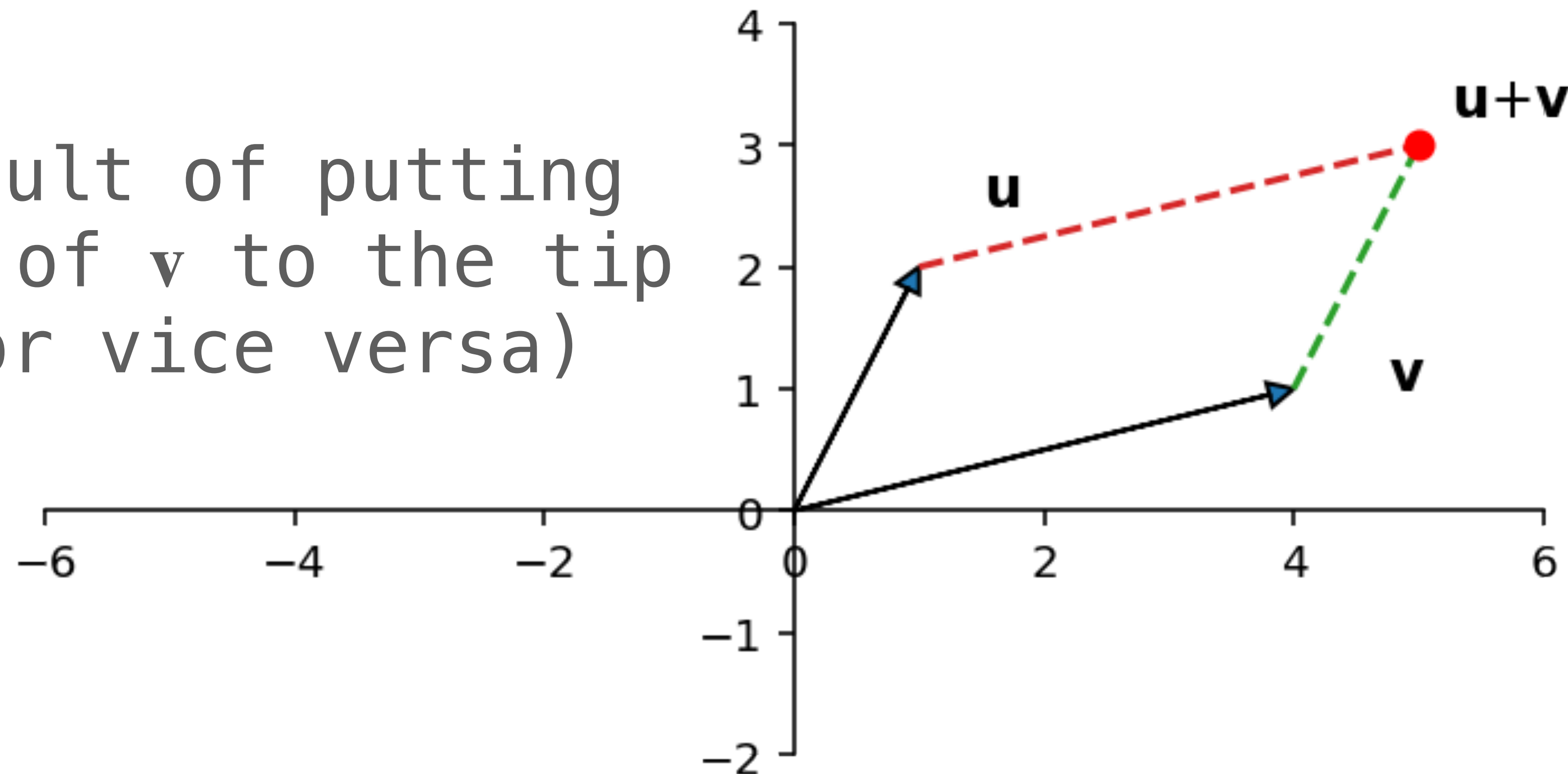
or the *tip-to-tail rule*



Vector Addition (Geometrically)

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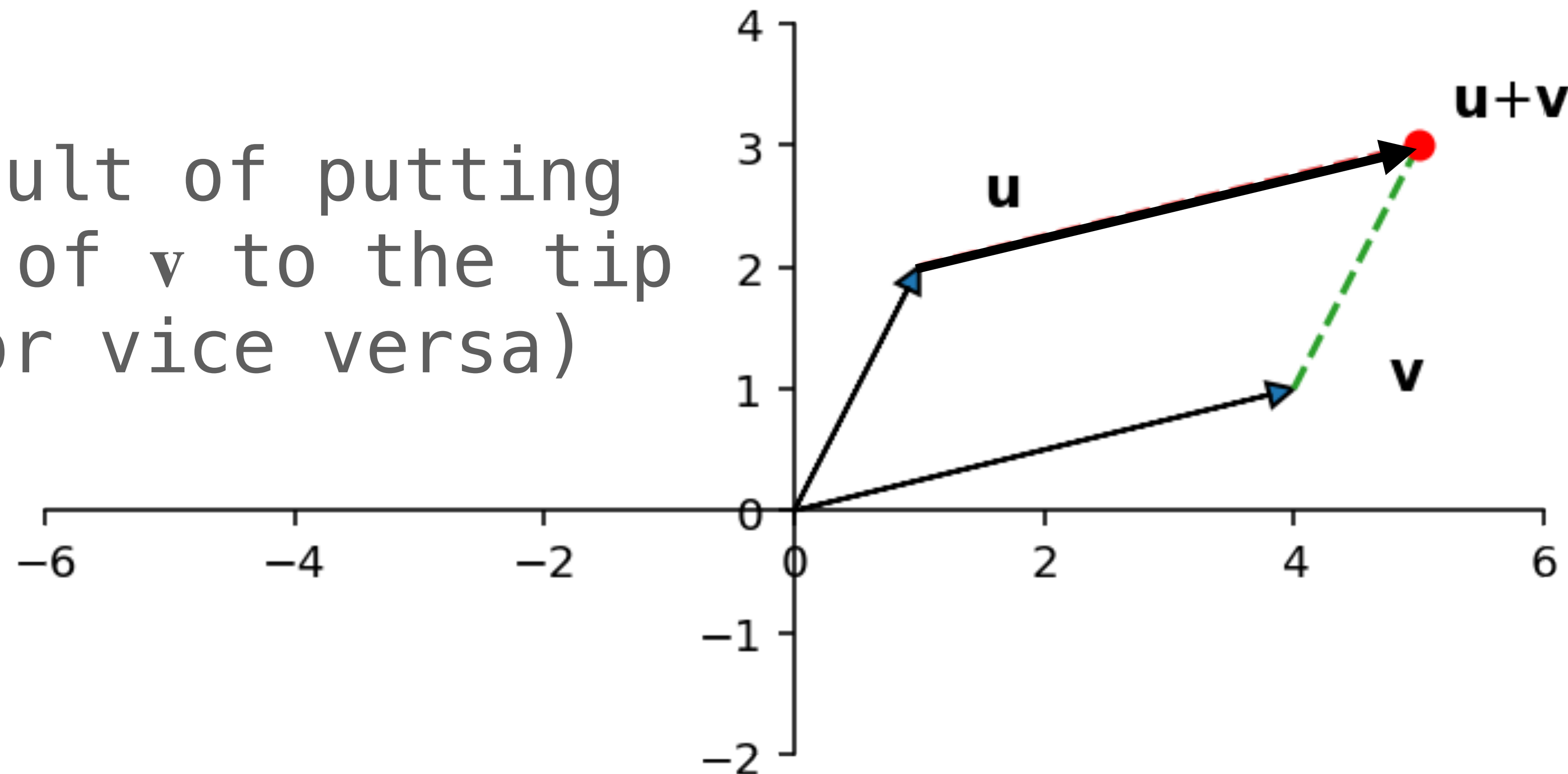
$\mathbf{u} + \mathbf{v}$ result of putting
the tail of $\mathbf{v}$ to the tip
of $\mathbf{u}$ (or vice versa)



Vector Addition (Geometrically)

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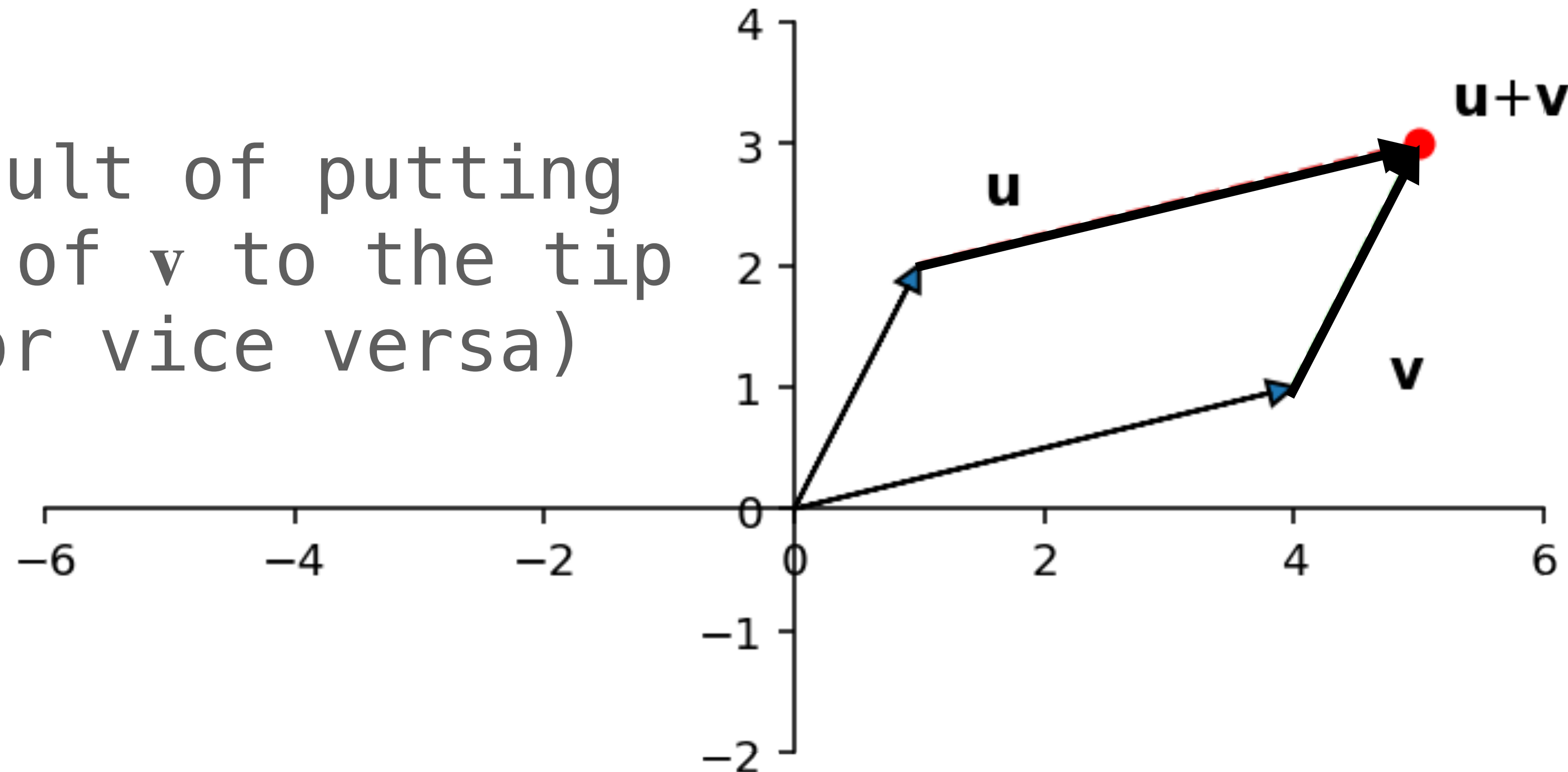
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demo
(from ILA)

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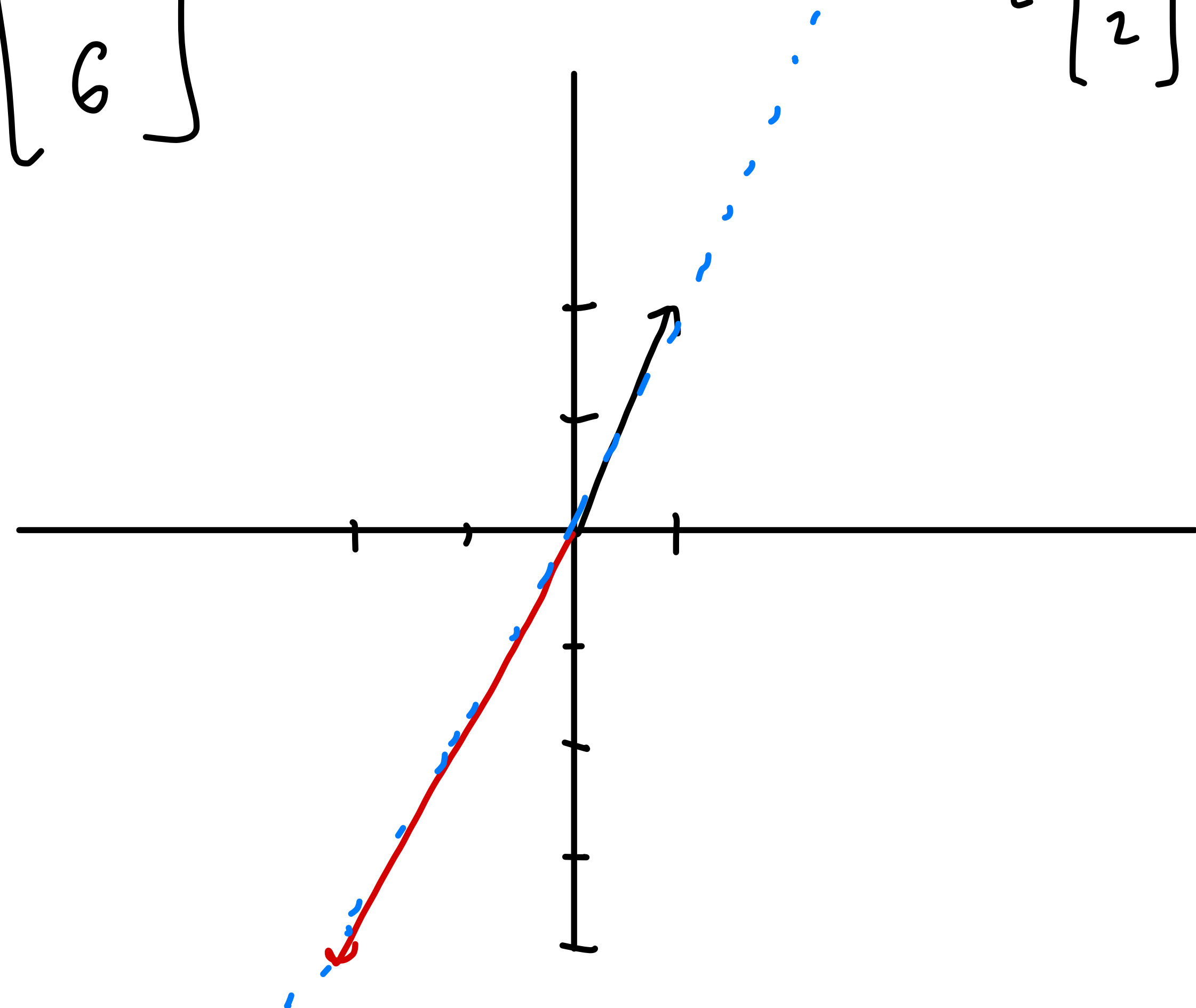
What properties do they need to satisfy?

Vector Scaling/Multiplication

$$2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2(5) \\ 2(3) \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

$$-2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \underline{\underline{\begin{bmatrix} -2 \\ -4 \end{bmatrix}}}$$

$$c \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}$$



Vector Scaling/Multiplication

scaling/multiplying a vector by a number means multiplying each of it's elements

$$a \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} ab_1 \\ ab_2 \\ \vdots \\ ab_n \end{bmatrix}$$

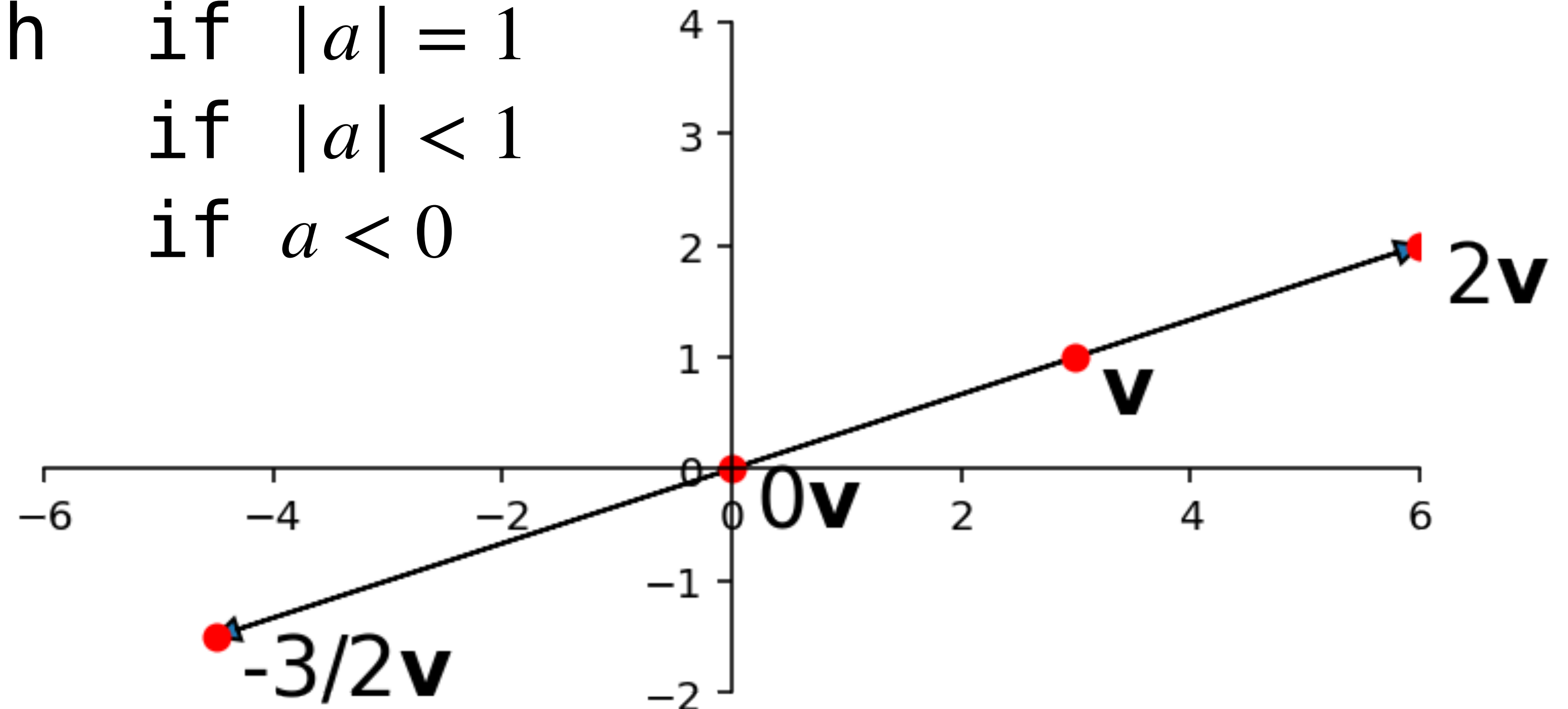
Vector Scaling/Multiplication (Example)

Scaling/multiplying a vector by a number means multiplying each of it's elements

$$3 \begin{bmatrix} 2 \\ 1 \\ 3.5 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \cdot 2 \\ 3 \cdot 1 \\ 3 \cdot 3.5 \\ 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 10.5 \\ 12 \end{bmatrix}$$

Vector Scaling (Geometrically)

longer	if $ a > 1$
the same length	if $ a = 1$
shorter	if $ a < 1$
reversed	if $a < 0$



demo
(from ILA)

Algebraic Properties

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and any real numbers c, d :

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$$

$$c(d\mathbf{u}) = (cd)\mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$$

$$1\mathbf{u} = \mathbf{u}$$

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these are requirements for any **vector space**
they matter more for *bizarre* vector spaces

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

Example "Proof"

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix} = \begin{bmatrix} v_1 + u_1 \\ \vdots \\ v_n + u_n \end{bmatrix} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

Question (Practice)

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

Compute the value of the above vector

Answer

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

In Sum

In Sum

we can add vectors

In Sum

we can add vectors

we can scale vectors

In Sum

we can add vectors

we can scale vectors

What vectors can we make with these operations?

Linear Combinations

Linear Combinations

Df. All possible sums / scalings of $v_1, \dots, v_n$

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 \dots c_n \vec{v}_n$$

Linear Combinations

Df. a *linear combination* of vectors

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$$

is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

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weights

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Df. a *linear combination* of vectors

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Looks suspiciously like
a linear equation...

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

weights

Linear Combinations (Example)

$$3 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} - \begin{bmatrix} -3 \\ 4 \\ 2 \\ 0 \end{bmatrix}$$

Definition. a *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are in $\mathbb{R}$

demo
(from ILA)

The Fundamental Concern

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3 \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Can $\mathbf{u}$ be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$?

That is, are there weights $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots \alpha_n \mathbf{v}_n = \mathbf{u}$?

Why is this fundamental?

I'm going to ask that you suspend your disbelief...

Why is this fundamental?

I'm going to ask that you suspend your disbelief...

For now, how do we solve this problem?

Vector Equations and Linear Systems

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

We want the weights...

The Fundamental Connection

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

We want the weights...

What if we write them as unknowns?

Some Symbol Pushing...

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x_1 \\ -2x_1 \\ -5x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

add

equations

$$\begin{bmatrix} x_1 + 2x_2 \\ -2x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

$$\begin{aligned} x_1 + 2x_2 &= 7 \\ -2x_1 + 5x_2 &= 4 \\ -5x_1 + 6x_2 &= -3 \end{aligned}$$

linear system

Some Symbol Pushing...

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$\begin{bmatrix} x_1 \\ (-2)x_1 \\ (-5)x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 5x_2 \\ 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$\begin{bmatrix} x_1 + 2x_2 \\ (-2)x_1 + 5x_2 \\ -5x_1 + 6x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

Some Symbol Pushing...

$$x_1 + 2x_2 = 7$$

$$(-2)x_1 + 5x_2 = 4$$

$$-5x_1 + 6x_2 = -3$$

Some Symbol Pushing...

$$x_1 + 2x_2 = 7$$

$$(-2)x_1 + 5x_2 = 4$$

$$-5x_1 + 6x_2 = -3$$

we get a system
of linear
equations we
know how to
solve

General Vector Equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{1m} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{21} \\ \vdots \\ a_{2m} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{nm} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

General Vector Equations

$$\begin{bmatrix} a_{11}x_1 \\ a_{21}x_1 \\ \vdots \\ a_{1m}x_1 \end{bmatrix} + \begin{bmatrix} a_{21}x_2 \\ a_{21}x_2 \\ \vdots \\ a_{2m}x_1 \end{bmatrix} + \dots + \begin{bmatrix} a_{n1}x_n \\ a_{n2}x_n \\ \vdots \\ a_{nm}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

by vector scaling

General Vector Equations

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

by vector addition

General Vector Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

by vector equality

The Fundamental Connection

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system of linear equations

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*This is our big
perspective shift
of the day*

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vector equation

How To: Linear Combination Problems

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Q. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$?

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$$\left[\begin{array}{c|c|c|c|c} | & | & & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_n & \mathbf{b} \\ | & | & & | & | \end{array} \right]$$

this is notation for
building a matrix
out of column
vectors

A solution to this system is a set of weights to define $\mathbf{b}$ as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$

Question

Can $\begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$ be written as a linear combination of $\begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$?

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \text{RREF}$$

exercise

Answer

Spans

Spans

Df. The span of $\{v_1, \dots, v_n\}$ is all possible
lin. comb. of $v_1, \dots, v_n$.

$$\text{span} \{ \vec{v}_1, \dots, \vec{v}_n \} = \left\{ \underbrace{\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n}_{\text{lin. comb.}} : \underbrace{\alpha_1, \dots, \alpha_n}_{\text{weights}} \in \mathbb{R} \right\}$$

ex:

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right\} \quad \begin{bmatrix} 10 \\ 7 \end{bmatrix} = 10 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - (1.5) \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$
$$\begin{bmatrix} 10 \\ 7 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right\} \quad = \begin{bmatrix} 10 \\ 10 \end{bmatrix} - \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix} - \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 7 \end{bmatrix}$$

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read: $\mathbf{u}$ is an element of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$



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demo
(from ILA)

Span of One Vector

Span of Two Vector

Spans (Geometrically)

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for one vector

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$$\text{span}\{\mathbf{v}\} = \{\alpha\mathbf{v} : \alpha \in \mathbb{R}\}$$

this is **all scalar multiple of $\mathbf{v}$**

the span of one vector is a **line**

Spans (Geometrically)

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the span of **two** vectors can be a **plane**

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the span of **two** vectors can be a **plane**

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!!IMPORTANT!!

In all cases they pass through the origin

demo
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you know how to do this now

Example

$$\text{Is } \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix} \text{ in span } \left\{ \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} \right\} ?$$

Question (Conceptual)

What does it mean geometrically if $\mathbf{b} \notin \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$?

demo
(from ILA)

How To: Inconsistency and Spans

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There is **no way** to write $\mathbf{b}$ as a linear combination

Example

*Find a vector **not** in* $\text{span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 3 \end{bmatrix} \right\}$

Summary

Vectors are fundamental objects

We can think of them as the **columns** of a linear system

We can **scale** them and **add** them together

They can **span** spaces which represent **hyperplanes**